

Foundation (Jalapeno)

- Q1.** A swimmer is training 15 meters below the surface (-15). What is the absolute value of the swimmer's depth?
- (A) -15
(B) 0
(C) 15
(D) -30
(E) 30
- Q2.** In a golf tournament, a score of 5 under par is written as -5 . Which expression represents the distance of this score from par (0)?
- (A) $|-5|$
(B) $-|5|$
(C) $-(-5)$
(D) $|5 - 5|$
(E) $|0 - (-5)|$
- Q3.** Calculate the total magnitude of yardage change for a football team that had a play of -12 yards followed by a play of $+12$ yards: $|-12| + |12|$.
- (A) 0
(B) 12
(C) -24
(D) 24
(E) 144
- Q4.** Which of the following athletes has the greatest absolute deviation from their target weight if their differences are: Athlete A (-4 lbs), Athlete B ($+3$ lbs), Athlete C (-2 lbs), Athlete D ($+1$ lb)?
- (A) Athlete D
(B) Athlete C
(C) Athlete B
(D) Athlete A
(E) All are equal
- Q5.** Evaluate the expression: $-|-25|$ representing a mountain climber's descent.
- (A) 25
(B) -25
(C) 0
(D) 50
(E) -50
- Q6.** A coach uses a number line to track sprints. If a player is at position x and $|x| = 9$, which two positions could the player be at?
- (A) 0 and 9
(B) -9 and 0
(C) 9 and -9
(D) 4.5 and -4.5
(E) 18 and -18
- Q7.** Simplify the fitness score expression: $|-8 + 3|$.
- (A) 11
(B) -11
(C) -5
(D) 5
(E) 0
- Q8.** Compare the intensity levels of two workouts: Workout X ($|-20|$) and Workout Y ($|18|$). Which statement is true?
- (A) $|-20| < |18|$
(B) $|-20| = |18|$
(C) $|-20| > |18|$
(D) $-20 > 18$
(E) $|18| = -20$

Open Ended Questions (FRQ)

- Q9.** A referee stands at the 0 -yard line. A player runs to the -22 yard line. Write an absolute value expression to represent the distance the referee must walk to reach the player and solve it.
- Q10.** During a fitness test, a heart rate monitor shows a deviation of d from the resting rate. If $|d| = 14$, what are the two possible actual deviations? Explain what the absolute value represents in this context.

Chef's Tips

- Remind students that absolute value always results in a non-negative distance.
- Use the 'distance from zero' visualization on a physical number line in the gym.
- Ensure students distinguish between $-|x|$ and $|-x|$.